

NBA Athlete Value Predictor

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Abstract—Each professional basketball player’s contract-based salary reflects their perceived value and can be viewed as a quantitative measure of their impact, skill, and contribution in the NBA. Accurately estimating salary therefore provides a mechanism for comparing and ranking players according to their theoretical value, offering a data-driven perspective on player quality. In this study, we develop a nonlinear framework that leverages performance metrics and related factors to estimate player salaries. By modeling the relationship between on-court performance and compensation, we construct a system that enables players to be evaluated on a common scale of inferred value. Additionally, we examine the relative importance of different features in shaping these estimates, providing further insight into the attributes most closely associated with high-value players. Our approach reframes salary prediction as a tool for player evaluation, allowing for the systematic ranking of athletes based on their projected worth. This perspective highlights how statistical performance translates into perceived value, offering a quantitative foundation for comparing players across the league.

I. INTRODUCTION

REFERENCES

- [1] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [2] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.